

Date:	15 March 2026	Team ID:	SWTID-2026-6684
Project Name:	ComicCraft - AI Comic Story Creator using Gemini Models		
Maximum Marks:	5 Marks		

Document Purpose: This document evaluates the quality of the **ComicCraft** codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future AI comic generator developments.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Strict PEP 8 4-space indentation enforced in all Python scripts.
2	Proper File Structure	Files and folders are logically organized	Yes	Separated modules: routes/, services/, templates/, and static/.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names like panel_outline, image_prompt, comic_pdf.
4	Function/Method Names	Functions are descriptively named	Yes	Verb-based naming e.g., generate_outline(), build_pdf_layout().
5	Code Comments	Inline and block comments explain logic	Yes	Comprehensive docstrings and inline comments on AI pipeline handlers.
6	Modular Design	Code is split into reusable functions/modules	Yes	Independent pipeline wrappers for Gemini Flash, Gemini Pro, and FPDF.
7	No Redundant Code	Duplicate or unused code is removed	Yes	DRY principle applied across route handlers and layout helpers.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	FastAPI HTTPException and fallback triggers for external AI API timeouts.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Gemini Flash Connector (gemini_flash.py)	Python, Google Generative AI SDK	Used across 5-panel JSON story outline generation & prompt parsing.	High
2	Gemini Pro Text Orchestrator (gemini_pro.py)	Python, Google Generative AI SDK	Reused for dialogue scripting, narration, and caption text generation.	High
3	Diffusers Image Pipeline (image_generator.py)	Python, PyTorch, Diffusers	Reused to render visual images for each of the 5 comic panels.	High
4	FPDF Comic Builder (exporters.py)	Python, FPDF Library	Reused for exporting dynamic multi-panel PDF documents.	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5 / 5	Exceptional modular separation between FastAPI routing layer, Gemini service calls, image generation routines, and PDF compilation modules.
Readability	5 / 5	Self-documenting code with highly descriptive variable and function names. Adheres strictly to Python PEP 8 standards.
Reusability	4.8 / 5	API client scripts (`gemini_flash.py`, `gemini_pro.py`, `image_generator.py`) are fully decoupled and can be integrated into future AI applications easily.
Documentation / Comments	4.7 / 5	Comprehensive module-level docstrings, inline comments explaining API JSON structures, and complete Swagger OpenAPI documentation via FastAPI `/docs`.
Overall Score	4.9 / 5	Production-ready, highly maintainable, and clean codebase for AI Comic Creation.

Quality Summary Verification

The code for project **ComicCraft - AI Comic Story Creator using Gemini Models** (Team ID: SWTID-2026-6684) successfully satisfies all parameters of clean code layout, high readability, and modular reusability. External AI APIs and export functionality are abstractly wrapped to ensure long-term maintainability.